



Wide binary formation and evolution in star clusters revealed by N-body simulations

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Abstract

Binary systems are frequent and important part of multiple systems that can be observed. Among them, soft/wide binaries, with low binding energy and large semi-major axis separations, are frequent in the galactic field despite their disruption tendency due to perturbations. We perform N -body simulations to give more in depth look to the dynamic evolution of such wide binaries in star clusters. We find that wide binaries continuously dissolve, but there are large number of dynamic formations of new binaries and reformation events of previously primordial wide binaries that counteract disruptions. By accounting these processes, we propose analytical model of the wide binary evolution that give short and long term disruption timescales of wide binaries in star clusters. Moreover, we explain influence of different binary fraction and rotation on clusters with large number of soft binaries. We compare binary fraction of wide binaries with observations and find similarities with Orion Nebula Cluster. Moreover, we propose optimal age range for observations of wide binaries within open clusters. We also observe dynamical mass segregation of binaries from the outskirts to center parts of the cluster in the later stages, but this is primarily contributed by hard binaries.

Introduction

Majority of objects in space come in different types of stellar multiplicities. The most frequent stellar multiplicity are binaries, consisting of gravitationally bounded 2 star. Depending on their binding energy, they can be classified in 2 groups (Heggie, 1975):

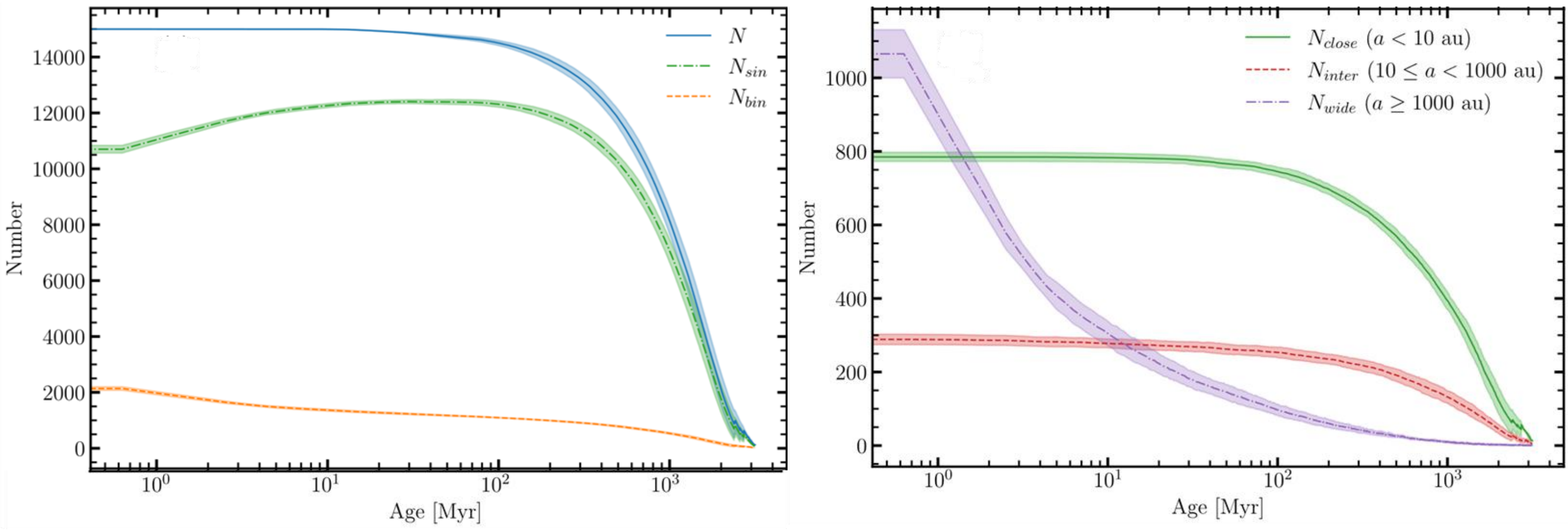
- **hard binaries** with high binding energy that tend to get harder
- **Soft binaries** with low binding energy that tend to get softer

Soft/wide binaries can have large semi-major axis separation and small binding energy which makes them disrupt until the relaxation timescale (Heggie, 1975).

Despite disruptions in early star cluster evolution, wide binaries are frequently observed in the field. El-Badry et al. (2018) found more than 53 000 candidates with projected separation between 50 to 50 000 au within 200 pc of the Sun via Gaia DR2. Continuation study with 1.8 million binary candidates from Gaia eDR3 show 99% alignment chance for binaries <30 000 au separation and high chance for binaries with separation up to 1 pc El-badry et al. (2023). Past studies suggested that wide binaries that are observed in the field are dynamically formed binaries that outlived their parent clusters (Kouwenhoven et al., 2010). Moreover, there are observations of wide binaries in the Orion Nebula cluster (ONC) with wide binary fraction of ~5% (Jerabkova et al., 2019). Other open clusters (OC) such as Alpha Per, the Pleiades, and Praesepe show wide binary fraction of ~2.1–3% (Deacon & Kraus, 2020). Observed wide binaries show chemical homogeneity that suggests common origins of pairs, possibly such star clusters, and non-random dynamical pairing (Hawkins et al., 2020). Moreover, some studies even suggest that wide binaries could be dominant source of stellar collisions in our galaxy and it could influence the formation of compact interacting binaries and merger products (Kaib & Raymond, 2014).

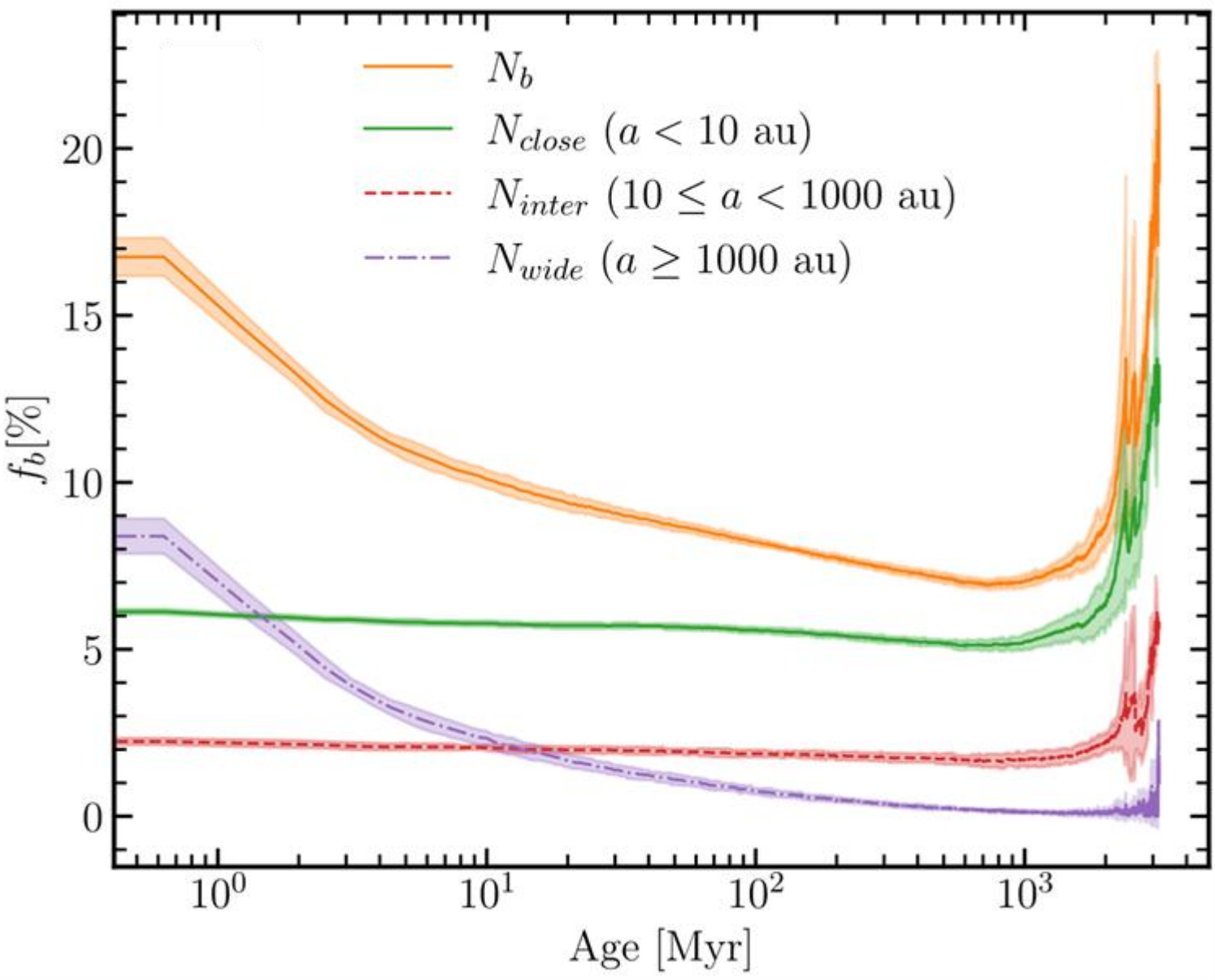
Results

From the start, 70–80% of wide binaries disrupt in the beginning and thus only 40% of total binaries are within the system. Wide binaries specifically rapidly decrease in the early stages.

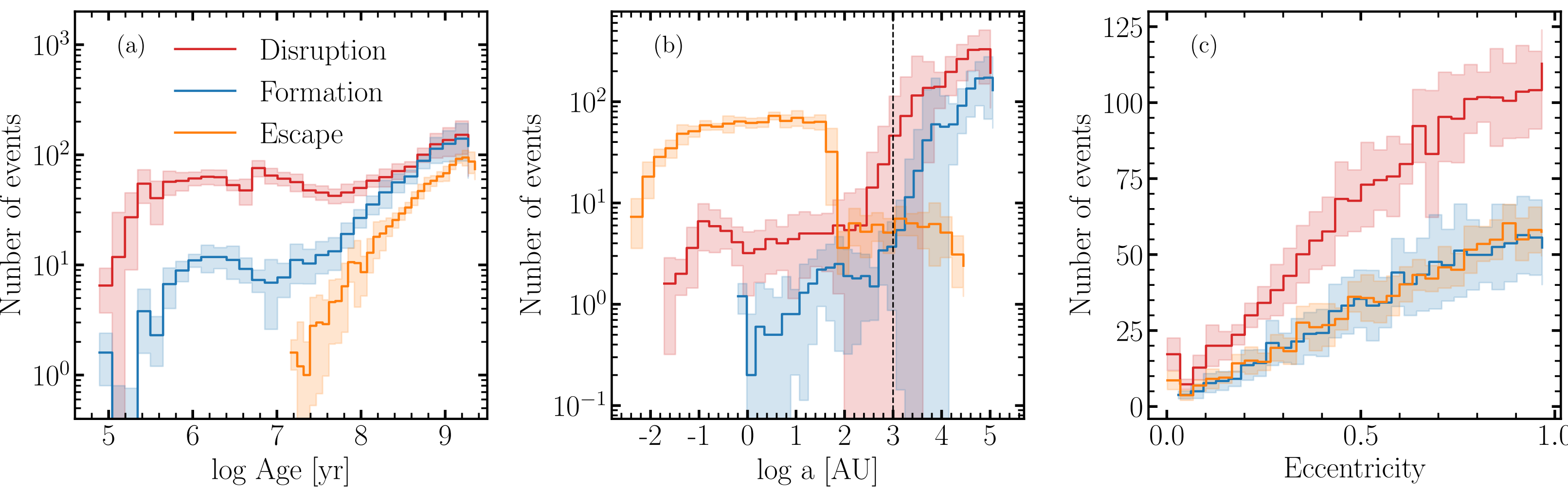


We also look at the binary fraction, f_b , as a function of time for different types of binaries.

Cluster	Age [Myr]	$f_{b,obs}$ [%]	$f_{b,sim}$ [%]
Orion Nebula cluster	2.5	5	4.5
Alpha Per	85	2	0.8
Praesepe	790	2.8	0.2
Pleiades	125	1.9	0.7



In order to have better understanding of the ongoing processes, we log the status change of binaries in simulations to track happened events. We classify events into 3 groups: *disruption*, *formation*, and *escape*. It is seen that disruption and decrease of wide binaries follows with dynamic formations of new wide binaries that somewhat compensate for disruptions.



Methodology

We use NBODY6++GPU, a high-performance GPU-accelerated code for gravitational N -body simulations (Kamlah et al., 2022b). NBODY6++GPU is a successor of the direct force integration N -body codes originally written by Sverre Aarseth (Aarseth et al., 2008).

The description of models can be seen in Tab. 1. There are 13 N -body simulations with different parameters. We use *Mclust* (Kamlah et al. 2022b) for creation of initial models and the output was used to run all the simulations. For showcase of results, we use *N15k_10hb_40sb* models.

Table 1
N-body simulation descriptions

Category	Name	Rands	N_{obj}	N	$f_{b,hard}$ [%]	$f_{b,soft}$ [%]	$f_{b,tot}$ [%]	ω_0
Binary influence (BI)	N12.5k_05hb_20sb	10	10k	12.5k	5	20	25	0.0
	N15k_10hb_40sb	10	10k	15k	10	40	50	0.0
	N17.5k_15hb_60sb	10	10k	17.5k	15	60	75	0.0
	N20k_20hb_80sb	10	10k	20k	20	80	100	0.0
Rotation	N15k_10hb_40sb_w0.03	5	10k	15k	10	40	50	0.3
	N15k_10hb_40sb_w0.06	5	10k	15k	10	40	50	0.6
	N15k_10hb_40sb_w0.09	5	10k	15k	10	40	50	0.9
	N15k_10hb_40sb_w0.12	5	10k	15k	10	40	50	1.2
	N15k_10hb_40sb_w0.18	5	10k	15k	10	40	50	1.8
BI on rotation	N12.5k_05hb_20sb_w0.12	5	10k	12.5k	5	20	25	1.2
	N15k_10hb_40sb_w0.12	5	10k	15k	10	40	50	1.2
	N17.5k_15hb_60sb_w0.12	5	10k	17.5k	15	60	75	1.2
	N20k_20hb_80sb_w0.12	5	10k	20k	20	80	100	1.2

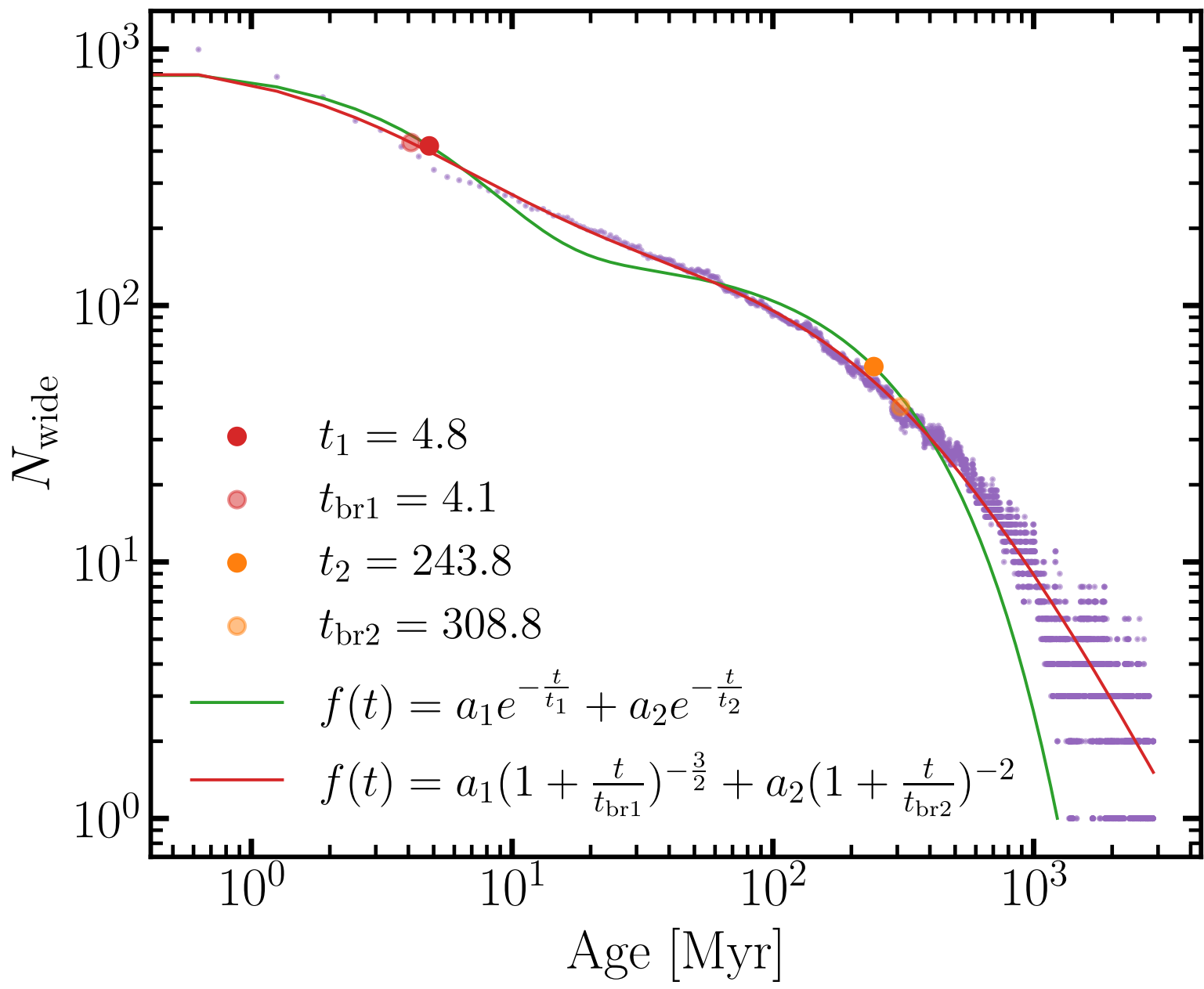
By considering disruptions and formations of wide binaries within star cluster, we propose analytical model of wide binary evolution in star clusters:

$$f(t) = a_1 \exp\left(-\frac{t}{t_1}\right) + a_2 \exp\left(-\frac{t}{t_2}\right)$$

where
 t_1 - short disruption timescale
 t_2 - long disruption timescale

$$f(t) = a_1 \left(1 + \frac{t}{t_{br1}}\right)^{-1.5} + a_2 \left(1 + \frac{t}{t_{br2}}\right)^{-2}$$

Where
 t_{br1} - early disruption break timescale
 t_{br2} - long disruption break timescale



Example of the fitting and values of t_1 and t_2 for all the models. t_1 and t_2 are seen to be shorter in the models with high binary fraction and rotation. These clusters are denser initially and it increases perturbations that make wide binaries disrupt.

Conclusion

We perform N -body simulations with high soft binary fraction and found various results.

- We've looked in depth at evolution of wide binaries and see a lot of predicted disruptions, but also dynamic formations of wide binaries in simulations
- By looking at disruptions and formations, we propose wide binary evolution model with short- and long-term disruption timescales.
- We compare wide binary fraction to observations and see high similarity with Orion Nebula Cluster and some discrepancies with Alpha Per, Praesepe and Pleiades.
- From the wide binaries within our simulations, we claim that most amount of wide binaries in open star clusters could be observed until 20 Myr.

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